

NYAH BENTLEY

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Broken LCD (SLM On)

For the damaged LCD (Figures 11-25), these effects are further amplified. Existing defects interact with the imposed patterns, producing exaggerated peaks and valleys in the reconstructed surface. As a result, defects appear significantly more pronounced compared to the unmodulated case, demonstrating that structured illumination enhances both real and artificial features.

Additionally, variations in pattern spatial frequency influence the appearance of the reconstruction. Higher-frequency patterns produce more localized and rapidly varying features, while lower-frequency patterns result in smoother, wave-like structures. These results confirm that the observed depth variations are not solely due to physical surface geometry but are strongly influenced by intensity gradients introduced by the display.

Checkerboard board circles and squares

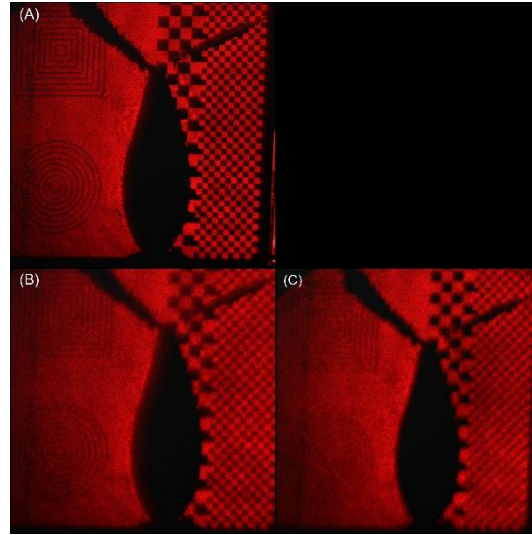


Figure 11 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying a checkerboard pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (−z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

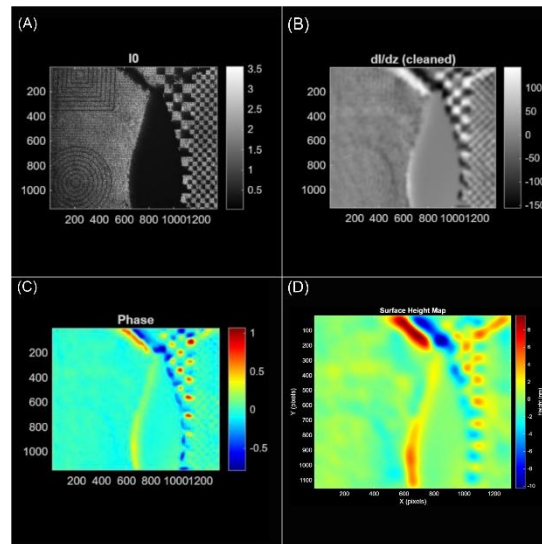


Figure 12 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I/\partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

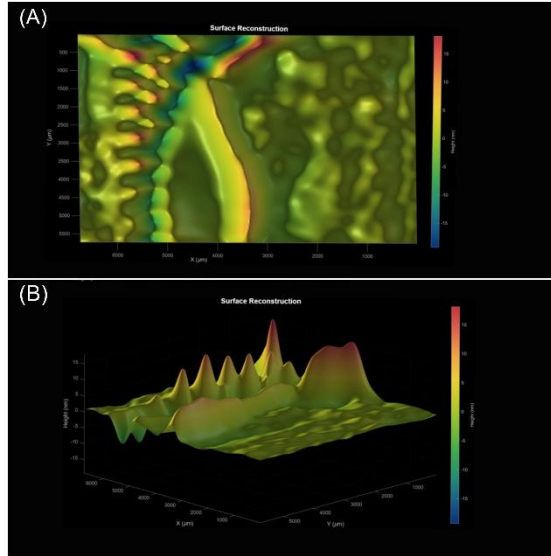


Figure 13 shows the 3d reconstruction of the checked damaged lcd. The damages show to be more intense than the checked patterns. (A) top view (B) Side view

Squares on the right side

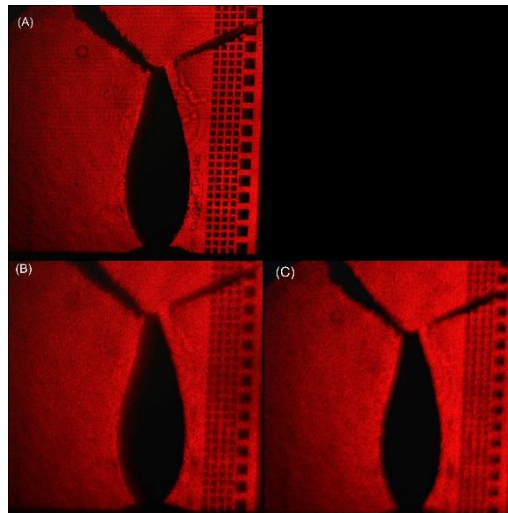


Figure 14 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Squares on the right-side pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

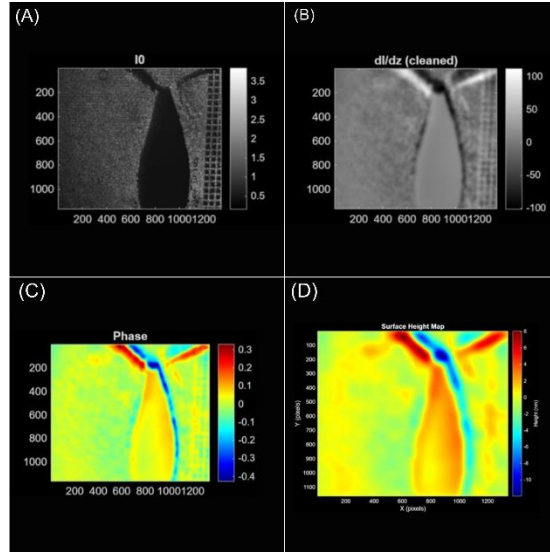


Figure 15 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

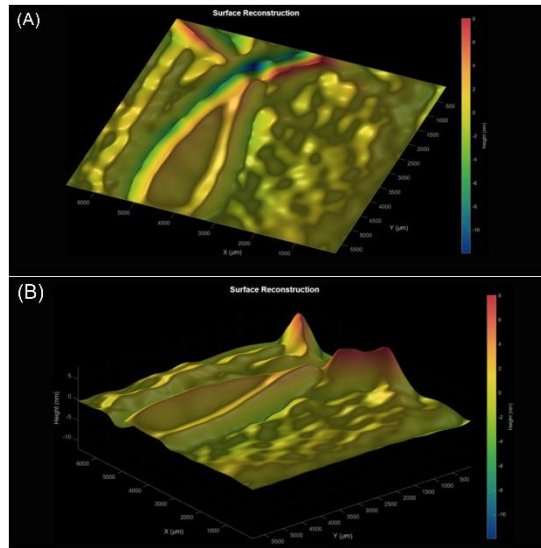


Figure 16 shows the 3d reconstruction of the checked damaged lcd. The damages show to be more intense than the checked patterns. (A) top view (B) Side view